RAG Learning

B1.4 Photosynthesis

OCR A GCSE Biology (J247) Gateway — Topic B1: Cell-level systems

1. What is Photosynthesis?

Photosynthesis is the process by which plants and algae use **light energy** (absorbed by chlorophyll) to convert **carbon dioxide** and **water** into **glucose** and **oxygen**. It is an **endothermic** reaction — it absorbs energy from the surroundings (light energy from the sun).

$$\text{Word: carbon dioxide} + \text{water} \rightarrow \text{glucose} + \text{oxygen (in light, with chlorophyll)}$$

$$\text{Symbol: } 6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$$

Photosynthesis takes place in the **chloroplasts** of plant cells (mainly in the **palisade cells** of the leaf), which contain the green pigment **chlorophyll**. Chlorophyll absorbs light energy — mainly red and blue wavelengths, reflecting green (which is why plants look green).

2. Leaf Structure — Adaptations

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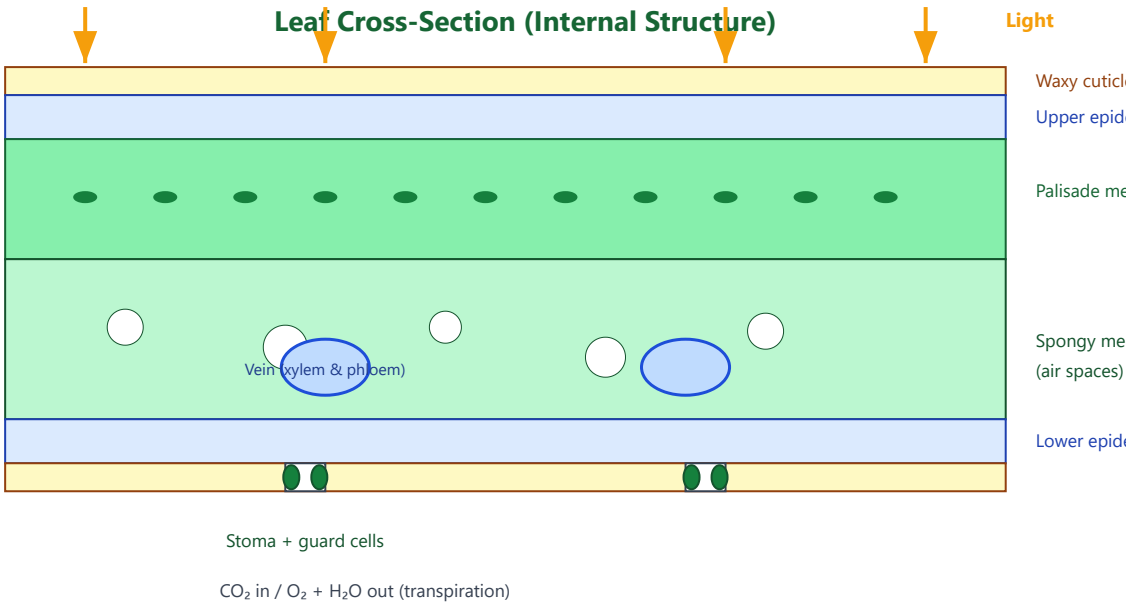
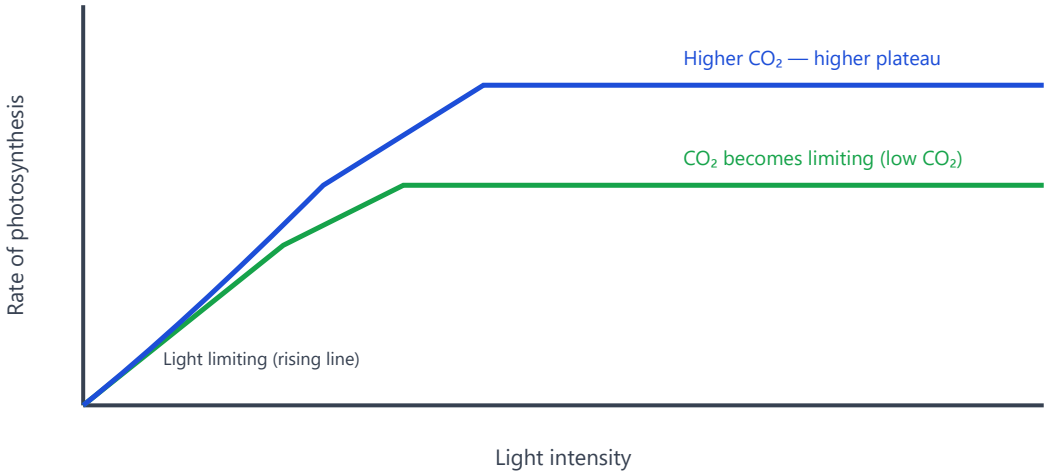


Adaptation	Function
Broad and flat	Large surface area to absorb light
Thin	Short diffusion distance for gases (CO_2 in, O_2 out)
Waxy cuticle (transparent)	Waterproof — reduces water loss but lets light pass through
Palisade mesophyll cells (column-shaped)	Packed with **chloroplasts** at the top — absorb the most light
Spongy mesophyll with air spaces	Allow CO_2 and O_2 to diffuse easily through the leaf
Stomata (mostly on lower surface)	Pores that allow gas exchange — opened/closed by guard cells
Veins (xylem and phloem)	Xylem brings water; phloem carries glucose away to other parts of the plant

3. Limiting Factors of Photosynthesis

A **limiting factor** is a factor that is in **shortest supply** and therefore **limits the rate** of photosynthesis. The three main limiting factors are **light intensity**, **carbon dioxide concentration** and **temperature**.

Effect of Light Intensity on Rate of Photosynthesis



Factor	Effect
Light intensity	As light increases, rate increases (more energy for the reaction). At high light intensity, another factor becomes limiting and the rate plateaus.
CO_2 concentration	As CO_2 increases, rate increases (more reactant). At high CO_2, another factor becomes limiting and rate plateaus.
Temperature	As temperature increases, rate increases — enzymes work faster. Above ~40 °C, enzymes denature and rate falls sharply.
Chlorophyll	Plants short of magnesium (or with leaf disease) make less chlorophyll, slowing the rate.

4. The Inverse Square Law

Inverse square law: Light intensity decreases as the *square* of the distance from the source. Doubling the distance gives ¼ of the light intensity.

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$$\text{Light intensity} \propto 1 / (\text{distance})^2$$

Worked example

A light source is moved from 10 cm to 20 cm from a plant. By what factor does the rate of photosynthesis decrease (assuming light is the limiting factor)?

1. Distance is doubled (×2), so light intensity decreases by $2^2 = 4$.
2. Light intensity becomes 1/4. Rate decreases by a factor of 4.

> **Exam Tip:** In an exam practical, this means moving the lamp from 10 cm to 30 cm gives 1/9 of the light. Always work out the SQUARE of the distance ratio.

5. Uses of Glucose in Plants

Glucose made by photosynthesis is used by the plant in several ways:

- **Respiration** — releases energy for cell processes
- Converted to **cellulose** for cell walls (growth)
- Combined with nitrogen ions absorbed from the soil to form **amino acids** (then proteins)
- Stored as **insoluble starch** (in leaves and storage organs like potatoes)
- Stored as **oils/lipids** in seeds (e.g. sunflowers)

6. Required Practical Link

Linked PAG: **investigating the rate of photosynthesis** using *pondweed* (e.g. Cabomba or Elodea) and counting bubbles of O_2 produced per minute, OR by measuring the volume of O_2 collected. Lamp distance is varied to change light intensity. See PAG notes for full method.

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